

6	(a)	(i)	$R = \rho L / A$	B1	
		(ii)	strain = $\Delta L / L$	B1	
			either $\Delta R = \rho \Delta L / A$ or $R \propto L$ with ρ and A constant	B1	
			dividing, $\Delta R / R = \Delta L / L$	A0	[3]
	(b)		Young modulus = stress / strain	C1	
			strain = $72.0 / (1.20 \times 10^{-7} \times 2.10 \times 10^{11})$	C1	
			= 2.86×10^{-3} (allow 1/350	A1	
			$\Delta R = 2.86 \times 10^{-3} \times 4.17 = 1.19 \times 10^{-2} \Omega$	A1	
			answer given to 3 sig. fig	B1	[5]
7	(a)		both measure (energy / work) / charge	B1	